

TactiLDM for VLA: Learning Unified Visual-Text-Tactile Representations to Empower Vision-Language-Action Policies

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Abstract—Integrating tactile perception into Vision-Language-Action (VLA) models is essential for achieving human-level dexterity and robust physical interaction in robotics. In this paper, we propose TactiLDM, a novel framework designed to learn unified visual-text-tactile representations within a Latent Diffusion Model (LDM) framework. Our approach employs a two-stage training strategy: 1) align tactile embeddings with vision and text modalities through contrastive learning; 2) utilize a conditional latent diffusion process, capturing both high-level semantics and intrinsic physical attributes. On Touch and Go (TAG) dataset, TactiLDM achieves 82.84% accuracy in tactile-only material classification, outperforming prior tactile encoders by over 15%. In zero-shot tactile task environment, π_0 policy with tactile modality integration achieves 4x increase in success rate and a 60% improvement in interaction stability. These results pave the way for more capable multimodal robotic systems. Code of our framework can be found at <https://github.com/lzy001Yuki/TactiLDM>

Index Terms—Tactile Sensing, Multimodal Representation Learning, Vision-Language-Action Models

I. INTRODUCTION

Learning robust and semantic-rich representations from multimodal sensory inputs is a cornerstone of embodied intelligence, enabling robots with information-rich interaction with the physical world and precise control on dexterous contact-rich manipulation tasks effectively. [1], [2] Recent advances in vision-language-action (VLA) models have pushed this direction forward by unifying perception, instruction understanding and action generation within a single architectural framework. [3], [4] Built upon this unified formulation, current-stage VLAs typically make their progress on emphasizing language ability and capability [3], rearranging VLA system centering a SOTA core model [5], or focusing on CoT reasoning as a reinforcement learning goal [6]–[8], etc.. Despite their architectural differences, they overwhelmingly remain vision-dominant, with VLMs serving as the finetuning model base [5], [9], [10] and visual input serving as the primary perceptual modality for grounding actions.

Yet as VLAs evolve, there is a central challenge for contact-rich manipulations: visual input alone cannot fully support

contact-rich manipulation. [11], [12] Among the available sensory channels in real-time robot action tasks, vision and touch offer highly complementary information: vision provides global scene layout, object geometry and semantic identity, whereas touch perception is capable of capturing localized contact geometry, material properties, frictional forces [13], and fine-grained interaction dynamics such as vibrations.

However, current tactile integration in action models remains constrained by sparsity and scarcity of data. This is because tactile and action signals are intrinsically temporally sparse [14], [15]. For a model to learn to effectively leverage touch, one in practice either collects specialized datasets that simultaneously include tactile and action data or relies on reinforcement learning to discover effective policies through trial and error. However, the latter approach is often impractical due to extremely sparse reward signals and prohibitively high training costs. This has led to an implicit consensus in implementation: although tactile sensing is valuable, remain a supplementary modality and not consume substantial training resources.

Despite significant progress in visual representation learning [16], [17], tactile perception application long remains under-explored in VLA models. Recent works in VLA models have attempted to bridge vision and touch via contrastive learning [18], masked modeling [19], or joint embedding alignment [20]. However, tactile signals—though commonly represented as images—are inherently low in high-level semantic content compared to natural vision. As a result, when processed by off-the-shelf visual encoders (e.g. ResNet [21]), tactile inputs yield representations that underperform significantly on semantic tasks compared to vision or text inputs.

Being in the situation of scarcity and sparsity of tactile-action data, we believe that future advances in multimodal capture and large-scale robot interaction pipelines will dramatically expand the availability of vision-touch and touch-action datasets. Guided by this outlook, our work is designed not merely to compensate for present limitations, but also to bring some enlightenment for next-generation vision-tactile-

language-action (VTLA) models and systems that can fully exploit rich tactile feedback. Our work introduce TactiLDM, a tactile-specific encoder within latent diffusion model framework.

Our main contribution is threefold:

- TactiLDM, a tactile encoder training framework with extensive model structure, highly adaptive alignment method and strong semantic ability.
- The tactile encoder of TactiLDM is easily swapable, and it integrates seamlessly with diverse vision or text encoders in contrastive learning with negligible performance loss, demonstrating robust implicit physical encoding capability.
- A low-cost, zero-shot tactile extension for VLA models via TactiLDM, which enables visuo-tactile fusion and improves action reasoning using only vision-tactile synchronized data—without requiring tactile-action supervision or reinforcement learning.

The rest of the paper is organized as follows. Section II gives a brief review of related works on tactile perception encoding and zero-shot modality extension. Section III describes the structure of TactiLDM in details. We publish our experimental setup, experiment result and evaluation in simulation environment as a proof for our zero-shot paradigm in Section IV. Finally we conclude our results, limitations and outlooks of related fields.

II. RELATED WORK

A. Touch Representation Encoding

Touch representation (ordinarily image-based, from Gel-Sight or similar sensors [22], [23]) encoding methodology determines the recognition and classification ability of an encoder. Embodied intelligence, such as VLAs, use *touch-as-vision* paradigm in majority, treat tactile data as visual inputs by directly applying pretrained vision encoders [9], [24], whose choices include ResNet [21] and SigLIP [25], [26], or pretrained vision-language models (VLMs). This paradigm makes the greatest convenience but lacks specific tactile understanding without adaptation refining e.g. LoRA. Variants [27], [28] leverage tactile-language models (TLMs) such as Octopi [29] and LLaMa [30] to generate language descriptions of tactile input.

Individual Tactile Encoder paradigm is popular as well. Our zero-shot extension experiment also use this paradigm. Typical structures usage include MLP [31], CNN [23], convolutional autoencoders [32] and ViTs [33], [34]. On novel methodology and systems, 3D-ViTac [35] proposes a real-world point cloud rebuilding system, while AnyTouch [36] employed a two-stage training framework to unify static and dynamic features. Encoders that aim on generalization on heterogeneous data, most famously T3 [23], routes specific encoders and project the features into a shared attention chunk. The model structure we choose and the training methodology design are distinguishing points, as open source tactile datasets [37]–[40] are relatively scarce but comprehensive on material classes.

B. Modality Extension of VLA

Vision-language-action (VLA) frameworks have traditionally unified vision and language for robotics [41], and recent works seek to extend this unification with additional physical sensory modalities such as audio [42] and touch. Tactile sensing has become the mainstream choice due to its strong complementarity to vision in contact-rich manipulation. Recent research that evolves VLA to vision-tactile-language-action models (VTLAs) converges on several methodological directions:

- *Semantic Alignment* between tactile signals and pre-trained vision-language representations, typically through contrastive or language-grounding objectives, allowing inheriting advanced semantic priors and few-shot transfer. [27] Our experiments give a zero-shot proof for such effectiveness, and TactiLDM provides a low-cost adaptive semantic alignment framework.
- *Cross-modal, Multi-level and late-fusion* methods add transformer-based fusion blocks or cross-attention layers to the VLA policy core, updating the shared latent space. [31] The additional modules require finetuning with tactile-action trajectories, yet improve the performance better on tasks [43] specified by the training data. [44]

III. METHOD

In the phase of tactile encoder design, our approach consists of two training stages within a Latent Diffusion Model (LDM) framework, aiming to learn a semantic-rich tactile representation aligned with visual and textual modalities.

1) *Stage 1: Tactile encoder with Multimodal Contrastive Learning*: In the first stage, we separately train a CNN encoder and a convolutional autoencoder for tactile image inputs, which are provided in RGB format. Let $\mathbf{x}_t \in \mathbb{R}^{H \times W \times 3}$ denote a tactile image. The encoder $E_t(\cdot)$ maps $\mathbf{x}_t$ to a latent representation $\mathbf{z}_t = E_t(\mathbf{x}_t) \in \mathbb{R}^d$.

To align the tactile modality with vision and language, we employ contrastive learning between the tactile embedding $\mathbf{z}_t$ and representations from other modalities:

- Vision: A pre-trained SigLIP [25] model encodes a vision image input $\mathbf{x}_v$ into $\mathbf{z}_v = \text{SigLIP}(\mathbf{x}_v) \in \mathbb{R}^d$.
- Text: A pre-trained T5 [45] encoder processes a label text input $\mathbf{c}$ into $\mathbf{z}_c = \text{T5}(\mathbf{c}) \in \mathbb{R}^d$.
- Tactile: An encoder encodes a tactile image input $\mathbf{x}_t$ into a latent representation via $E_t(\cdot)$, which is then projected to the shared embedding space by a learnable tactile projector $P_t(\cdot)$, yielding $\tilde{\mathbf{z}}_t = P_t(E_t(\mathbf{x}_t)) \in \mathbb{R}^d$. We consider two variants: $E_t^{CNN}(\cdot)$, employs a convolutional neural network as the tactile encoder and $(E_t^{AE}(\cdot), D_t^{AE}(\cdot))$, employs a convolutional autoencoder.

In both variants, we aim to minimize the InfoNCE loss [46] across all modality pairs:

$$\mathcal{L}_{\text{InfoNCE}}(\mathbf{z}_t, \mathbf{z}_c) = -\log \frac{\exp(\mathbf{z}_t^\top \mathbf{z}_v / \tau)}{\sum_{\mathbf{z}' \in \mathcal{N}_v} \exp(\mathbf{z}_t^\top \mathbf{z}' / \tau)} - \log \frac{\exp(\mathbf{z}_t^\top \mathbf{z}_c / \tau)}{\sum_{\mathbf{z}' \in \mathcal{N}_c} \exp(\mathbf{z}_t^\top \mathbf{z}' / \tau)}. \quad (1)$$

where τ is a temperature hyperparameter, and $\mathcal{N}_v, \mathcal{N}_c$ denote negative samples from the vision and text modalities, respectively. All projected embeddings are L2-normalized: $\hat{\mathbf{z}} = \mathbf{z} / \|\mathbf{z}\|_2$.

For any two modalities A and B with normalized embeddings $\{\hat{\mathbf{z}}_i^A\}_{i=1}^N$ and $\{\hat{\mathbf{z}}_i^B\}_{i=1}^N$ from a batch of N aligned samples, we define the contrastive loss as:

$$\mathcal{L}_{A \leftrightarrow B} = \frac{1}{2} \left(\frac{1}{N} \sum_{i=1}^N \mathcal{L}_{\text{infoNCE}}(\hat{\mathbf{z}}_i^A, \hat{\mathbf{z}}_i^B) \right), \quad (2)$$

The total contrastive loss differs between the two tactile encoder variants.

For CNN-encoder $E_t^{CNN}(\cdot)$, we leverage all three pairwise alignments to maximize cross-modal consistency

$$\mathcal{L}_{\text{cont}} = \mathcal{L}_{\text{tactile} \leftrightarrow \text{vision}} + \mathcal{L}_{\text{tactile} \leftrightarrow \text{text}} + \mathcal{L}_{\text{vision} \leftrightarrow \text{text}}. \quad (3)$$

For autoencoder $(E_t^{AE}(\cdot), D_t^{AE}(\cdot))$, we combine reconstruction loss of the autoencoder with $\mathcal{L}_{\text{tactile} \leftrightarrow \text{vision}}$ and $\mathcal{L}_{\text{tactile} \leftrightarrow \text{text}}$.

$(E_t^{AE}(\cdot), D_t^{AE}(\cdot))$ is trained end-to-end with a standard reconstruction loss $\mathcal{L}_{\text{recon}} = \|\mathbf{x}_t - D_t^{AE}(E_t^{AE}(\mathbf{x}_t))\|_2^2$, yielding the total loss function:

$$\mathcal{L}_1 = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{tactile} \leftrightarrow \text{vision}} + \mathcal{L}_{\text{tactile} \leftrightarrow \text{text}}, \quad (4)$$

. Upon convergence, we freeze the encoder $E_t^{AE}(\cdot)$ and discard the decoder for subsequent stages.

2) *Stage 2: Latent Diffusion Model with Multimodal Conditioning*: In the second stage, we adopt and freeze the encoder $E_t(\cdot)$ from Stage 1 to embed tactile inputs into a latent space suitable for diffusion modeling. Given a tactile image $\mathbf{x}_t$, we obtain its latent representation,

$$\mathbf{z}_0 = E_t(\mathbf{x}_t). \quad (5)$$

We then apply the standard forward diffusion process over T timesteps:

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t} \mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (6)$$

where $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ controls noise scheduling.

The core of Stage 2 is a U-Net denoising network $\mathcal{U}_\theta$ that predicts the noise $\boldsymbol{\epsilon}$ given a noisy latent $\mathbf{z}_t$, timestep t , and multimodal conditioning tokens. Specifically, we form the conditioning vector by concatenating the frozen visual and textual embeddings:

$$\mathbf{c}_{\text{cond}} = [\mathbf{z}_v; \mathbf{z}_c] \in \mathbb{R}^{2d}. \quad (7)$$

During denoising, $\mathbf{c}_{\text{cond}}$ is added into cross-attention layers. The training objective is the standard diffusion loss:

$$\mathcal{L}_2 = \mathbb{E}_{\mathbf{z}_0, t, \boldsymbol{\epsilon}} [\|\boldsymbol{\epsilon} - \mathcal{U}_\theta(\mathbf{z}_t, t, \mathbf{c}_{\text{cond}})\|_2^2]. \quad (8)$$

Depending on the architecture of the Stage-1 tactile encoder, we distinguish two variants of this model: **TactiLDM**, which uses a CNN backbone $E_t^{CNN}(\cdot)$, and **auto-TactiLDM**, which adopts an autoencoder-based encoder $E_t^{AE}(\cdot)$. Both yield a multimodal-aligned tactile representation that enables high-fidelity generation and facilitates downstream tasks.

IV. EXPERIMENT

A. Linear Probe

Datasets. We conduct our experiments on the *Touch and Go* (TAG) dataset [37], a large-scale, human-collected benchmark. TAG includes paired visual frames and GelSight tactile images, along with object category annotations, making it suitable for probing tasks.

Objective. To evaluate the quality of the tactile representation learned in *Visual-Text-Tactile Unified Representation Learning* Phase, we adopt linear probing method: freeze pre-trained tactile encoder $E_t(\cdot)$ and only train a lightweight linear classifier on top of its fixed representations. High performance on a downstream classification task under this setting strongly suggests that $E_t(\cdot)$ has captured high-level.

To prevent data leakage during representation learning, we mask out text modality in the conditional token. This ensures that the learned tactile representations do not encode ground-truth labels.

Baselines. To demonstrate the effectiveness of our **TactiLDM** and **auto-TactiLDM**, we compare the performance of our learned tactile representation with a vision-based encoder **Siglip** [25], and three tactile encoder **UniTouch** [47], **ViT-Lens-2** [48], **OmniBind** [49].

Experiment Results. We evaluate the quality of our learned representations under two distinct settings: (1) a *Tactile-Only* setting, to measure the intrinsic discriminative ability of the tactile encoder, and (2) a *Tactile-Vision Fused* setting, to assess how tactile features complement visual information when combined.

Tactile-Only Performance. Table I summarizes the linear probing accuracy using tactile image inputs only. We observe that both variants of our proposed framework, **TactiLDM** and **auto-TactiLDM**, significantly outperform existing baselines on downstream classification tasks, where for example, in the detailed material classification task, **auto-TactiLDM** achieves an accuracy of 82.84%, surpassing the strong vision-language baseline SigLIP by over 15% and the unified tactile baseline UniTouch by over 20%.

This highlights the advantages of our method in bridging the modality gap. While SigLIP possesses strong semantic priors, it struggles to generalize to the tactile modality without explicit alignment. Additionally, while UniTouch and ViT-Lens-2 are designed for tactile data, our results suggest that the integration of Latent Diffusion Model creates a more semantically dense representation than contrastive learning alone.

Tactile-Vision Fused Performance. To investigate whether our learned tactile features provide information complementary to vision, we conduct an evaluation using both modalities (Table II). We concatenate the frozen embeddings from our tactile encoder with the visual embeddings from SigLIP, and for Siglip baseline, we directly concatenate both tactile and vision embeddings from itself.

While SigLIP performs strongly due to the rich semantic context, the addition of our tactile representation pushes performance to near-saturation. **TactiLDM** achieves 97.44%

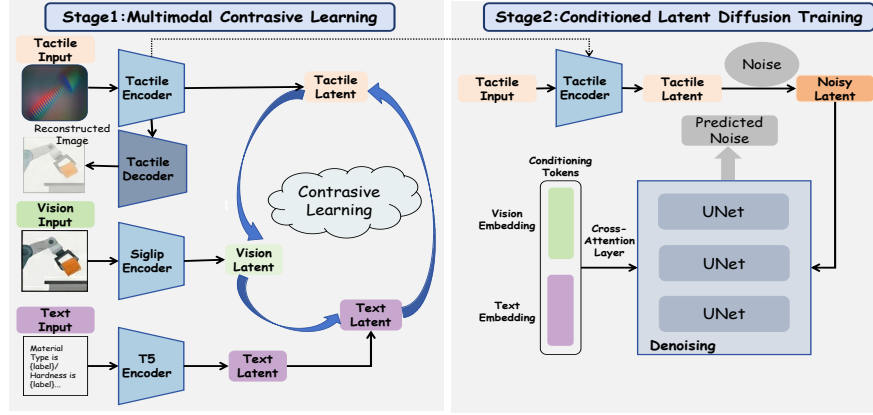


Fig. 1. Training process for auto-TactiLDM. For TactiLDM, discard tactile decoder design at Stage 1.

on material classification and 99.34% on hardness estimation. This improvement demonstrates that our model captures physical properties that are imperceptible to vision alone, thereby effectively bridging the modality gap.

Implicit Physical Encoding. Beyond explicit classification benchmarks, we further investigate the intrinsic semantic ability of our model. We observe that when **TactiLDM** is only trained with ‘full’ labels which help classify material types, it achieves a success rate of **92.34%** on hardness estimation (in *Tactile Input Only Setting*). This demonstrates the model’s ability to *implicitly* encode fundamental material-level priors. This finding underscores the effectiveness of our Latent Diffusion framework in transferable physical representations that go beyond simple pattern matching.

TABLE I
LINEAR PROBING ACCURACY ON TAG (TACTILE INPUT ONLY)

Encoder	Touch and Go		
	Material	Rough	Hard
Siglip	67.19%	71.22%	73.75%
UniTouch	62.77%	64.23%	70.12%
ViT-Lens-2	64.09%	68.56%	-
OmniBind	67.45%	-	-
TactiLDM	81.18%	86.97%	94.68%
auto-TactiLDM	82.84%	87.03%	94.28%

Bold font denotes the best performance.

TABLE II
LINEAR PROBING ACCURACY ON TAG (TACTILE-VISION INPUT)

Encoder	Touch and Go		
	Material	Rough	Hard
Siglip	90.44%	92.58%	96.69%
TactiLDM	97.44%	98.89%	99.34%

B. VTLA Performance On Maniskill-Vitac 2025

Platform. We conduct our experiments on the **ManiSkill-ViTac 2025** challenge [50] simulation environment.

Our experimental setup employs the challenge’s focus on *Tactile-Vision Fusion Manipulation* (Track 2). An illustration of the simulation setup is shown in Figure 2, which depicts the peg-insertion task along with visual and tactile observations.

Baseline. We adopt π_0 [51] as baseline, which employs a novel flow matching architecture built upon a pre-trained Vision-Language Model, demonstrating strong zero-shot generalization and the ability to perform dexterous tasks.

Challenges and Approach. Current tactile simulation capabilities are limited. Most existing tactile datasets are collected from real-world hardware. In simulation environments including **ManiSkill-ViTac 2025** [50], there is a scarcity of trajectory demonstrations, where VLA models are confined to interacting with the environment only through Reinforcement Learning. This restricts the application of imitation learning.

Because of these challenges, we bypass the need for fine-tuning π_0 . Instead, we directly adopt π_0 inference by concatenating tactile observations into its input modality stream and evaluate it in a *zero-shot* setting.

Metrics. We employ the following metrics to quantify model’s performance:

- **Success Rate (SR):** The percentage of evaluation episodes that are completed successfully according to the task-specific success criteria.
- **Average Truncation Step (ATS):** The average timestep at which an episode is terminated. The platform simulates a soft, deformable tactile sensor. If the policy causes excessive deformation during action chunk, a `tactile_movement_too_large` error is triggered, causing immediate episode truncation. A higher ATS indicates the model effectively interprets tactile images and modulates its actions to maintain safe contact.

Experimental Results. The experimental results in Table III demonstrate that adding tactile observations into the input stream of π_0 without any fine-tuning leads to a substantial improvement in both success rate and interaction stability.

The success rate increases from 5% to 20%, representing a **4×** improvement in zero-shot task completion. This highlights

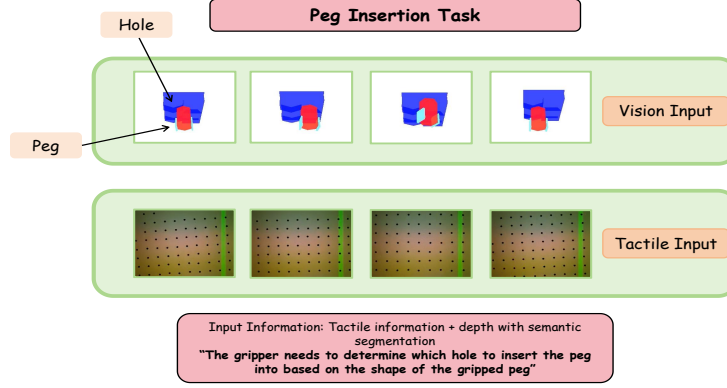


Fig. 2. Illustration for Peg Insertion Task on Maniskill-ViTac2025

the critical role of tactile feedback in tasks requiring precise physical interaction.

Moreover, the Average Truncation Step (ATS) rises from 155.71 to 249.98 steps. This +60% increase in interaction duration demonstrates that the integration of tactile data allows the robot to modulate its actions more effectively, maintaining safer contact and avoiding immediate simulation truncation.

TABLE III
ZERO-SHOT PERFORMANCE COMPARISON ON THE MANISKILL-ViTAC 2025 PEG INSERTION TASK. THE MAXIMUM EPISODE LENGTH IS 500 STEPS.

Metrics	π_0 (Baseline)	π_0 (+ Tactile Input)
SR $\uparrow$	5%	20%
ATS $\uparrow$	155.71	249.98

V. CONCLUSION

In this work, we introduced **TactiLDM**, a framework designed to bridge the gap between tactile, visual, and textual modalities through Latent Diffusion Models. By employing multimodal contrastive alignment followed by conditional latent diffusion, we successfully developed a tactile encoder capable of capturing both semantic categories and physical attributes.

Our experiments demonstrate that these representations excel in downstream classification tasks in both tactile-only setting ($\approx 20\%$ improvement) and tactile-vision fused setting ($\approx 5\%$ improvement). Furthermore, our zero-shot evaluation on the ManiSkill-ViTac 2025 challenge underscores the practical value of these representations for Vision-Language-Action models, leading to a **4x** increase in success rates for peg-insertion tasks. TactiLDM bridges the gap between generative modeling and embodied perception. Future work will explore the scaling of TactiLDM to larger, more diverse tactile datasets and its integration into end-to-end control policies.

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